Data Science

BSc Computer Science - Durham University

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1 Module Introduction

Data Science has the module code COMP2271 and is led by Professor Thomas Erlebach.

1.1 Assessment

This module is assessed through two pieces of summative coursework and two final exams, these can be seen in Table 1. Additional formatives are also shown in italics.

Title	Opening	Due	Feedback	Professor	Weight
Data Cleaning & Analytics	Week 11	-	11th Feb	Dr Igor Razgon	25%
Image Processing	15th Jan	2nd Mar	6th May	Dr George Koulieris	25%
Probability and Graphics	May Exam	-	-	Razgon & Koulieris	50%

Table 1: Assessments for Data Science.

Data Cleaning & **Analytics Benchtest Information**: The Data Cleaning & Analytics test in Week 11 is a closed book, multiple choice test, and will be held during practicals (mine is Friday 4pm).

- Closed-book assessment over one hour. Check information closer to the time.
- Comprised of 15 multiple-choice questions, split equally between: basic NumPy, basic Pandas, data cleaning, data analytics, and visualisation.

1.2 Timetable

This module is taught through two one-hour lectures and one two-hour computer class each week as seen below in Table 2.

Day	Time	Class Type	Location	Lecturer
Monday	12:00 - 13:00	Lecture	W103	Dr Igor Razgon
Friday	10:00 - 11:00	Lecture	W103	Dr Igor Razgon
Friday	16:00 - 18:00	Computer Class	CC-0007	-

Table 2: Michaelmas Term Timetable for Data Science.

2 Probability and Statistics

This section is taught by Dr Igor Razgon. The recommended reading for this topic is:

- Probability and Statistics (Degroot, M.H. and Schervish, M.J., 2012).

2.1 Probability and Statistics: Basic Concepts

2.1.1 Counting Theory

Definition 2.1.1.1: The product rule states that, if a procedure can be broken down into a sequence of two tasks with n_1 ways to do the first task and n_2 ways to do the second task, then there are $n_1 n_2$ ways to complete the entire procedure.

Example 2.1.1.1: John wants to fly to Canada from Germany with a connecting flight through the UK. There are 3 flights from Germany to the UK, and 2 flights from the UK to Canada. How many possible flight combinations are there?

Solution: $3 \times 2 = 6$.

□

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Definition 2.1.1.2: The sum rule states that, if you consider two independent sub-tasks (we only have to do one), if the first can be done in n_1 ways, and the second can be done in n_2 ways, then the total number of ways to complete the task is $n_1 + n_2$.

Example 2.1.1.2: Mary wants to choose a student project from three lists: lecturer 1 offers 5 projects, lecturer 2 offers 2 projects, and lecturer 3 offers 10 projects. How many possible options for her?
Solution: $5 + 2 + 10 = 17$.

□

Definition 2.1.1.3: The inclusion-exclusion principle states that, if two tasks can be done at the same time (unlike the sum rule in which they are independent), then there is a possibility of overcounting since some ways may be counted twice. To counter this, we add up the number of ways to do each task, then subtract the number of ways to do both tasks.

$$|A \cup B| = (|A| + |B|) - (|A \cap B|) \tag{1}$$

$$|A \cup B \cup C| = (|A| + |B| + |C|) - (|A \cap B| + |A \cap C| + |B \cap C|) + (|A \cap B \cap C|) \tag{2}$$

Definition 2.1.1.4: An r-permutation of a set of r distinct objects is one ordered arrangement. We denote the number of r-permutations of a set with n elements in as $P(n, r)$.

$$P(n, r) = n \times (n-1) \times (n-2) \times ... \times (n-r+1) = \frac{n!}{(n-r)!} \tag{3}$$

Suppose that a set consists of n objects, of which n_1 are of one type (indistinguishable from one another), n_2 are of a second type, ..., and n_k are of a k_{th} type where $n = n_1 + n_2 + ... + n_k$. The number of permutations of the objects can be calculated with (4).

$$P(n; n_1, n_2, ..., n_k) = \frac{n!}{n_1! n_2! ... n_k!} \tag{4}$$

Definition 2.1.1.5: A r-combination is a counting of distinct objects in an unordered arrangement of length r. The number of r combinations of a set with n distinct elements is given by (5).

$$C(n, r) = \binom{n}{r} = \frac{n!}{(n-r)! \times r!} \tag{5}$$

Example 2.1.1.3: Assume elements A_1, A_2, A_3, A_4 are in a set. We want to choose 2 elements from them.
Solution:

- If the order of elements is important: $P(4, 2) = \frac{4!}{(4-2)!} = 12$.
- If the order of elements is not important: $C(4, 2) = \frac{4!}{(4-2)!2!} = 6$.

□

2.1.2 Probability Theory

Definition 2.1.2.1: Probability is the formalisation of uncertain events. It enables one to infer probabilities of interest based on assumptions and observations.

Definition 2.1.2.2: Statistics is the practice of the science of collecting and analysing data. It deals with experimental data and extracting knowledge from it.

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Definition 2.1.2.3: An experiment is a procedure that yields one of a given set of possible outcomes.

Definition 2.1.2.4: The sample space of the experiment is the set of possible outcomes.

Definition 2.1.2.5: An event is any subset of the sample space.

Theorem 2.1.2.1: The Theory of Probability states that, given a finite sample space of all outcomes S, assuming that all outcomes are equally likely, for an event set E where $E \subseteq S$, then the probability of occurrence of the event is given by (6).

$$P(E) = \frac{|E|}{|S|} \tag{6}$$

Proof: Proof omitted.

□

Definition 2.1.2.6: The axioms of probability are as follows:

- $P(E)$ is always between 0 and 1: $0 \leq P(E) \leq 1$.
- $P(S) = 1$.
- For disjoint events $E_1, E_2, ..., E_n$ (with no intersection), $P(E_1 \cup E_2 \cup ... \cup E_n) = \sum_{i=1}^{n} P(E_i)$.

Then, from the Axioms of Probability above, we can derive the following:

- **Definition 2.1.2.7**: The complement rule states that $P(\overline{E}) = P(S - E) = P(S) - P(E) = 1 - P(E)$.
- **Definition 2.1.2.8**: The impossible rule states that $P(\emptyset) = 0$.
- **Definition 2.1.2.9**: The general addition rule states that $P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$.
- If $E_1 \subseteq E_2$, then $P(E_1) \leq P(E_2)$.

Example 2.1.2.1: A sequence of 10 bits is randomly generated. What is the probability that at least 1 of these is 0?
Solution: Let E be an event in which all bits are 1.

$$P(E) = \frac{|E|}{|S|} = \frac{1}{2^{10}}$$

Then, by the complement rule, $P(\overline{E}) = P(1 - E)$ where $\overline{E}$ is the event in which at least one bit is not 1.

$$P(\overline{E}) = 1 - \frac{1}{1024} = \frac{1023}{1024}$$

□

2.2 Dependency and Bayes Theorem

2.2.1 Conditional Probability

The probability of event A, given that event B happens is given by:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \tag{7}$$

Therefore, the probability of both A and B occuring can be given by:

$$P(A \cap B) = P(A|B) \times P(B) \tag{8}$$

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Definition 2.2.1.1: Events can occur independently. The occurrence of one event is not dependent on the occurrence of another if, and only if:

$$P(A \cap B) = P(A) \times P(B) \tag{9}$$

Definition 2.2.1.2: $P(A \cap B)$ can be presented in two ways, either $P(A|B) \times P(B)$ or $P(B|A) \times P(A)$. Equating these, we can obtain Bayes' Theorem.

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)} \tag{10}$$

This can be rewritten by using variables, where $P(A|B)$ is the posterior, $P(A)$ is the prior, $P(B|A)$ is the likelihood, and $P(B)$ is the evidence.

$$posterior = \frac{likelihood \times prior}{evidence} \tag{11}$$

Suppose that $E_1, E_2, ..., E_n$ are mutually exclusive events such that:

$$\bigcup_{i=1}^{n} E_i = S \tag{12}$$

This means that one of the events **must** occur. Then, in any case, Bayes' Theorem states:

$$P(E_k|A) = \frac{P(E_k) \times P(A|E_k)}{\sum_{j=1}^{n} P(E_j) \times P(A|E_j)} \tag{13}$$

2.3 Discrete Probability Distributions

2.3.1 Random Variables

Definition 2.3.1.1: A random variable is a function that assigns a number to each outcome in the sample space of a random experiment. It is also known as a stochastic variable. They are usually denoted by a capital X or Y.

Definition 2.3.1.2: We say that X is a discrete random variable if X can take a finite number of k values $x_1, .x_2, ..., x_k$ and an infinite sequence of different values $x_1, x_2, ...$ which can be distinguished from one another.

Definition 2.3.1.3: The probability distribution of a random variable X describes the probabilities associated with each possible outcome of X. Let X be a discrete random variable. Suppose that the possible values it can assume are $x_1, x_2, ..., x_n$. Then, these values are associated with probabilities given by

$$P(X = x_k) = f(x_k), k = 1, 2, ..., n \tag{14}$$

This function f is known as the probability mass function (PMF).

2.3.2 Distribution Functions

Definition 2.3.2.1: The probability mass function (PMF) is function where the value is a probability and has the following characteristics:

- $f(x_k) \geq 0$.
- $\sum_{k=1}^{n} f(x_k) = 1$.

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The mathematical definition is that, for a discrete random variable X with possible values $x_1, x_2, ...$ the probability mass function is a function $f : \{x_1, x_2, ...\} \to [0, 1]$ satisfying $f(x_i) \geq 0$, $\sum_{k=1}^{\infty} f(x_i) = 1$, and $P(X = x_i) = f(x_i)$. They can be calculated by collecting all possible outcomes where $X = x$ and summing their probabilities for all x.

Definition 2.3.2.2: The cumulative distribution function (CDF) or distribution function for a random variable X is defined as

$$F(x) = P(X \leq x) = \sum_{x_i \leq x} P(X = x_i) = \sum_{x_i \leq x} f(x_i), \text{ where } x \in \mathbb{R} \tag{15}$$

Definition 2.3.2.3: The mean (μ) or expected value E of a function is the measure of dispersion or variability in a distribution.

Definition 2.3.2.4: The variance of a function is the measure of dispersion or variability in a distribution.

It is important to consider two laws when discussing random variables:

1. Functions of random variables are also random variables. Let X be a random variable, and $h : \mathbb{R} \to \mathbb{R}$ be an arbitrary function. Then $Y = h(X)$ is also a random variable.
2. **Definition 2.3.2.5**: The law of the unconscious statistician: Let X be a random variable with PMF f and $h : \mathbb{R} \to \mathbb{R}$ be an arbitrary function. Then $Y = h(X)$ is also a random variable. If X is discrete and takes values $\{x_1, x_2, ...\}$ then

$$E[Y] = E[h(x)] = \sum_i h(x_i) f(x_i) \tag{16}$$

These two laws result in the conclusion that one can perform arithmetic operations with random variables to get new ones, and the expectation and variance can be computed from them.

- $E(aX + b) = aE[X] + b$.
- $E[h(X)] = \sum_x h(x) f(x)$.
- $V(aX + b) = a^2 V(X)$.
- $V(X) = E(X^2) - [E(X)]^2$.

From this, $E[X]$ is named the first moment, and $E[X]^2$ is the second moment. The variance is also called the central second moment.

We can consider two random variables X and Y to be independent if, for sets A and B, we have

$$P(\{X \in A\} \cap \{Y \in B\}) \equiv P(X \in A, Y \in B) = P(X \in A) P(Y \in B) \tag{17}$$

For independent random variables only, we have

$$V(X_1 + X_2 + ... + X_n) = V(X_1) + V(X_2) + ... + V(X_n) \tag{18}$$

This equation holds for expectation E, whether the variables are independent or not.

2.3.3 Special Discrete Distributions

Definition 2.3.3.1: The discrete uniform distribution is the simplest discrete random variable. It assumes a finite number of possible values, each with equal probability. A random variable X has a discrete uniform distribution if each of the n values in its range $x_1, x_2, ..., x_n$ has equal probability. Suppose random variable X has uniform distribution. The probability mass function of X is given by ($X \sim Uni(n)$).

$$P(X = k) = f(k) = \frac{1}{n} \tag{19}$$

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If we assume X is a discete uniform distribution on the consecutive integers $a, a+1, a+2, ..., b$ for $a < b$, then:

$$\mu = E[X] = \frac{a+b}{2} \tag{20}$$

$$\sigma^2 = Var(X) = \frac{(b-a+1)^2 - 1}{12} \tag{21}$$

Definition 2.3.3.2: The most common discrete distributions are associated with Bernoulli trials. In a random experiment if each outcome consists of only two probabilities, such as in a toss of a coin, then the random experiment is called a Bernoulli trial. In a Bernoulli trial, let the possible events in each outcome be denoted by A and B. Then, $A \cap B = \emptyset$ and $A \cup B = S$. Let the occurrence of A be a success, and the occurrence of B be a fail and let the probability of A be p, then:

$$P(B) = P(\overline{A}) = 1 - P(A) = 1 - p = q \tag{22}$$

Let X be a random variable taking only two possible values $\{0, 1\}$ and let $p = P(X = 1)$. Then X is said to be a Bernoulli random variable with parameter p: $X \sim Ber(p)$, then:

$$E[X] = 0 \times P(X = 0) + 1 \times P(X = 1) = p \tag{23}$$

$$E[X^2] = 0^2 \times P(X = 0) + 1^2 \times P(X = 1) = p \tag{24}$$

$$V(X) = E[X^2] - (E[X])^2 = p - p^2 = p(1-p) \tag{25}$$

Taking (22) and (25), we get that $E[X] = p$ and $V(X) = pq$.

Definition 2.3.3.3: Consider a random experiment consisting of $n \in \mathbb{N}$ independent Bernoulli trials. If the probability of success for each trial equals p, $0 \leq p \leq 1$, and it remains the same throughout the experiment, then we say that the random variable X that equals the number of successful trials is a binomial random variable with parameters n and p. The probability mass function of X is given by $X \sim Bin(n, p)$.

$$P(X = k) = f(k) = \binom{n}{k} p^k (1-p)^{n-k}, k = 0, 1, ..., n \tag{26}$$

The binomial random variable is n independent repetitions of a Bernoulli variable, $X = X_1 + X_2 + ... + X_n$. So we can calculate

$$E[X] = E[X_1 + X_2 + ... + E_n] = E[X_1] + E[X_2] + ... + E[X_n] \tag{27}$$

$$V(X) = V[X_1 + X_2 + ... + X_n] = V(X_1) + V(X_2) + ... + V(X_n) \tag{28}$$

Therefore, $E[X] = np$ and $V(X) = np(1-p)$.

Definition 2.3.3.4: Consider a series of independent Bernoulli trials with constant probability of success p. Let the random variable X denote the number of trials until the first success. Then X is called a geometric random variable with parameter p: $X \sim Geom(p)$.

$$P(X = k) = f(k) = p(1-p)^{k-1}, k = 1, 2, ... \tag{29}$$

and $E[X] = \frac{1}{p}$ and $V(X) = \frac{1-p}{p^2}$.

Theorem 2.3.3.1: The geometric distribution has no memory of the past. This is called the memoryless property, meaning that if you have waited k trials already without seeing a success, you must wait m more trials to get success with probability of $p(1-p)^{m-1}$: $P(X = k + m | X > k) = P(X = m)$.

Proof:

$$P(X = k + m | X > k) = \frac{P(\{X = k + m\} \cap \{X > k\})}{P(\{X > k\})} \tag{30}$$

$$\frac{P(X = k + m)}{P(X > k)} = \frac{P(X = k + m)}{1 - P(X \geq k)} \tag{31}$$

$$= \frac{p(1-p)^{k+m-1}}{(1-p)^k} = p(1-p)^{m-1} = P(X = m) \tag{32}$$

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□

Definition 2.3.3.5: We can generalise the idea behind the geometric distribution - instead of waiting for the first success, we can wait until r successes. Consider a series of independent Bernoulli trials with constant success probability p. Let the random variable X denote the number of trials until $r \in \mathbb{N}$ successes are attained. Then X is called a negative binomial random variable with parameter p and r: $X \sim NegBin(p, r)$.

$$P(X = k) = f(k) = \begin{cases} 0, \text{ if } k < r \\ \binom{k-1}{r-1} p^r (1-p)^{k-r}, \text{ if } k \geq r \end{cases} \quad , \text{ where } k \in \mathbb{N} \tag{33}$$

Then $E[X] = \frac{r}{p}$ and $V(X) = r\frac{1-p}{p^2}$.

Definition 2.3.3.6: Suppose that you want to model the number of customers entering a store over the period of one day. To do this we make two assumptions: the customers behave independently from one another and the expected number of customers per unit of time is constant λ.

Breaking the day into four-hour intervals gives six pieces ($n = 6$), so the expected number of customers for each interval is $\frac{\lambda}{n} = \frac{\lambda}{6}$. If the intervals are really small, then the number of customers is either 0 or 1 with very high probability. Then the number of customers in each interval is approximately a Bernoulli random variable with $p = \frac{\lambda}{n}$, they are also all independent.

Now look at getting k arrivals in a $Bin(n, p)$, where the number of bins n grows towards $+\infty$. In this case, the probability of a single event will be $p = \frac{\lambda}{n}$.

A Poisson random variable has a distribution that shows how many times an event is likely to happen within a specific period of time. X is called a Poisson random variable with parameter $\lambda > 0$. It is the number of events in a Poisson process, $X \sim Poisson(\lambda)$.

$$P(X = k) = f(k) = \frac{e^{-\lambda}\lambda^k}{k!}, k = 0, 1, 2, ... \tag{34}$$

and $E[X] = \lambda$ and $V(X) = \lambda$.

2.4 Continuous Probability Distributions

2.4.1 Continuous Random Variables

Definition 2.4.1.1: A continuous random variable is a random variable such that the probability of any one specific value is unknown, and thus is treated as zero. Instead, we calculate the probability of the value being in a given interval.

Definition 2.4.1.2: In general, continuous random variables are described by a probability density function. It maps a probability of a range of events from the sample space to a function. The probability density function of a continuous random variable X is denoted f_X or simply f.

$$P(a \leq X \leq b) = \int_a^b f_X(x)dx \tag{35}$$

We define the probability density function $f(x)$ for a continuous random variable X such that:

- $f(x) \geq 0, \forall x \in \mathbb{R}$.
- $\int_{-\infty}^{+\infty} f(x)dx = 1$.
- $P(a \leq X \leq b) = \int_a^b f(x)dx, \forall (a, b) | (a, b) \in \mathbb{R}^2, a \leq b$.

Definition 2.4.1.3: Similarly, we can define the continuous cumulative distribution function $F(x) : \mathbb{R} \to [0, 1]$ of a continuous random variable X where $x \in \mathbb{R}$ as

$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)dt \tag{36}$$

As in the discrete case, a continuous CDF has the following properties:

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- $F(x)$ is non-decreasing: if $x \leq y$ then $F(x) \leq F(y)$.
- $0 \leq F(x) \leq 1$: $\lim_{x\to-\infty} F(x) = 0$, $\lim_{x\to+\infty} F(x) = 1$.
- For $a \leq b$ we have: $P(a \leq X \leq b) = F(b) - F(a)$.

If $F(x)$ is the cumulative distribution function of a continuous random variable X, then its derivative is the probability distribution function $f(x)$.

$$f(x) = F'(x) = \frac{d}{dx}F(x) \tag{37}$$

Example 2.4.1.1: A continuous random variable X has the density function (38). Find the value of the constant c.

$$f(x) = \frac{c}{x^2+1} \text{ where } -\infty < x < +\infty \tag{38}$$

Solution: From Definition 2.4.1.3, we know that

$$F(+\infty) = P(X \leq +\infty) = \int_{-\infty}^{+\infty} f(t)dt = 1 \tag{39}$$

Substituting in $f(x)$, we get

$$\int_{-\infty}^{+\infty} \frac{c}{x^2+1}dx = [\arctan(x)]_{-\infty}^{+\infty} = 1 \tag{40}$$

$$c \lim_{x\to\infty} [(\arctan(x) - \arctan(-x))]_{-\infty}^{+\infty} = c\left[\left(\frac{\pi}{2}\right) - \left(-\frac{\pi}{2}\right)\right] = 1 \tag{41}$$

This gives us that, $1 = c\pi$, giving $c = \frac{1}{\pi}$.

□

2.4.2 Mean and Variance

Going back to the continuous random variable X, we can define its mean and variance as values to summarise its distribution such that,

$$\mu = E[X] = \int_{-\infty}^{+\infty} xf(x)dx \tag{42}$$

and

$$\sigma^2 = V(X) = E\left[(X-\mu)^2\right] = \int_{-\infty}^{+\infty} (x-\mu)^2 f(x)dx \tag{43}$$

As for discrete random variables, mean and variance have the following properties:

- $E[aX + b] = aE[X] + b$.
- $V(aX + b) = a^2V(X)$.
- $\sqrt{V(aX+b)} = |a|\sqrt{V(X)}$.

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2.4.3 Specific Continuous Distributions

Definition 2.4.3.1: The continuous uniform distribution sets all samples in a range as having the same probability. Formally, let X be a continuous random variable with density function

$$f(x) = \begin{cases} \frac{1}{b-a}, \text{ if } a \leq x \leq b \\ 0, \text{ otherwise} \end{cases} \tag{44}$$

We have that the mean and variance of this distribution are $E(X) = \frac{b+a}{2}$ and $V(X) = \frac{(b-a)^2}{12}$.

Definition 2.4.3.2: The exponential distribution is used to model the time elapsing between events. The probability of occurrence of an event after a unit of time. It has a parameter λ which is the rate parameter and is the average number of events occuring in a period of one time unit.

Formally, a random variable X is called an exponential random variable with parameter $\lambda > 0$ (denoted by $X \sim Exp(\lambda)$) when the probability density function is

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, \text{ if } x \geq 0 \\ 0, \text{ otherwise} \end{cases} \tag{45}$$

and

$$F(x) = \begin{cases} 1 - e^{-\lambda x}, \text{ if } x \geq 0 \\ 0, \textit{ otherwise} \end{cases} \tag{46}$$

We can then define the mean and variance as $E(X) = \frac{1}{\lambda}$ and $V(X) = \frac{1}{\lambda^2}$.

2.5 Normal (Gaussian) Distributions

2.5.1 Introduction to the Normal Distribution

Definition 2.5.1.1: A normal distribution usually represents the sum of many independent random variables, and almost always looks like a bell curve. These bell curves can have different centres and spreads depending on the mean μ and variance σ^2 of the random variables.

Formally, a random variable X is said to have normal distribution with parameters mean $-\infty < \mu < +\infty$ and variance $\sigma^2, \sigma > 0$, written as $X \sim \mathcal{N}(\mu, \sigma^2)$. The probability density function is defined as:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \tag{47}$$

with $E(X) = \mu$ and $Var(X) = \sigma^2$.

2.5.2 The Standard Normal Distribution

Definition 2.5.2.1: The standard normal distribution is a special case of the normal distribution defined formally as: A random variable Z with probability density function as follows is called a standard normal random variable:

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \tag{48}$$

This is a normal random variable with parameters $\mu = 0$ and $\sigma^2 = 1$. The corresponding cumulative distribution function is denoted as $\Phi(z) = P(Z \leq z)$. This means that any normal distribution X can be converted into a standard normal distribution Z using the formula $Z = \frac{X-\mu}{\sigma}$. This is called the z-score or Z-score, and can be used to compare scores from distributions with different scales, or to determine the relationship between one score and the rest. Before using the normal approximation $P(X \leq x)$ should be changed to $P(X \leq x + 0.5)$.

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2.5.3 Normal Approximations

When you have a binomial distribution where n is large and p is the middle-of-the-road, then the binomial distribution tends to look like a normal distribution.

$$X \sim \text{Bin}(n,p) \Rightarrow P(X \leq x) \approx \Phi\left(\frac{x-np}{\sqrt{npq}}\right) \tag{49}$$

This approximation is generally good for $np > 5$ and $n(1-p) > 5$.

If X is a Poisson random variable with $E(X) = \lambda$ and $V(X) = \lambda$, then

$$X \sim \text{Poisson}(\lambda) \Rightarrow P(X \leq x) \approx \Phi\left(\frac{x-\lambda}{\sqrt{\lambda}}\right) \tag{50}$$

The same continuity correction used for the binomial distribution can also be applied. The approximation is good for $\lambda > 5$.

2.6 Multiple Random Variables

Definition 2.6.0.1: Distributions with two variables are called bivariate distributions, and those with more than two are multivariate distributions.

2.6.1 Bivariate Distributions

Definition 2.6.1.1: The probability mass function of a single discrete random variable X specifies how much probability mass if placed on each possible X value. The joint PMF of two discrete variables X and Y describes how much probability mass is placed on each possible pair of values (x, y).

Definition 2.6.1.2: The notion of join probability distribution can be formally defined as: If X and Y are discrete random variables, the function given by $f(x,y) = P(X = x, Y = y)$ for each pair of values (x, y) within the range of X and Y is called the joint probability distribution (joint probability mass function) of X and Y.

This joint PMF of X and Y inside the range R and S respectively has the following properties:

- $f_{XY}(x,y) \geq 0$.
- $\sum_{x \in R} \sum_{y \in S} f_{XY}(x,y) = 1$.
- $f_{XY}(x,y) = P(X = x, Y = y) \equiv P(\{X = x\} \cup \{Y = y\})$.

Definition 2.6.1.3: Then, we can extend the above definition to the Joint Cumulative Function as follows: If X and Y are discrete random variables, the joint CDF is the function given by:

$$F(x,y) = P(X \leq x, Y \leq y) = \sum_{s \leq x} \sum_{t \leq y} f(s,t) \text{ for } -\infty < x, y < +\infty \tag{51}$$

2.6.2 Splitting the Distribution

Definition 2.6.2.1: Although a bivariate probability distribution describes the joint probability, we may still need to explore X and Y separately. This is done through marginal probability distributions. Basically, the goal is to remove the other variable from the probability distribution.

Let X and Y be two discrete random variables taking values in range R and S respectively with joint probability mass function $f_{XY}(x,y)$. Then, the corresponding marginal probability mass functions are given by:

$$f_X(x) = P(X = x) = \sum_{y \in S} P(X = x, Y = y) = \sum_{y \in S} f_{XY}(x,y) \tag{52}$$

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$$f_Y(y) = P(Y = y) = \sum_{x \in R} P(X = x, Y = y) = \sum_{x \in R} f_{XY}(x, y) \tag{53}$$

Definition 2.6.2.2: For conditional distributions, let X and Y be two discrete random variables taking values in range R and S respectively with joint probability mass function $f_{XY}(x, y)$. Assume $f_X(x) > 0$ (meaning $X = x$ is possible), then:

$$f_{Y|X=x}(y) = P(Y = y|X = x) = \frac{P(X = x, Y = y)}{P(X = x)} = \frac{f_{XY}(x, y)}{f_X(x)} \tag{54}$$

Similarly, if $f_Y(y) > 0$:

$$f_{X|Y=y}(x) = P(X = x|Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)} = \frac{f_{XY}(x, y)}{f_Y(y)} \tag{55}$$

If X and Y are discrete random variables, then:

- $\sum_{y \in Y} f_{Y|X=x}(y) = 1$.
- For any set B:

$$P(Y \in B|X = x) = \sum_{y \in B} f_{Y|X=x}(y) \tag{56}$$

2.6.3 Continuous Variables

If we consider X taking a value in $[a, b]$ and Y a value in $[c, d]$:

- (X, Y) takes a value in $[a, b] \times [c, d]$.
- If the joint probability density function is $f_{XY}(x, y)$ then:

$$P(a \leq X \leq b, c \leq Y \leq d) = \int_a^b \int_c^d f_{XY}(x, y)dydx \tag{57}$$

- $f_{XY}(x, y)dxdy$ can be interpreted as the probability of lying in a small rectangle.

Let X and Y be two continuous random variables. The joint PDF of X and Y has the following properties:

- $f_{XY}(x, y) \geq 0$.
- $\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f_{XY}(x, y)dydx = 1$.
- For any set R in the R^2 plane:

$$P((X, Y) \in R) = \int \int_R f_{XY}(x, y)dydx \tag{58}$$

Let X and Y be two continuous random variables with joint PDF $f_{XY}(x, y)$. Then the corresponding marginal PMFs are given by:

$$f_X(x) = \int f_{XY}(x, y)dy \tag{59}$$

$$f_Y(y) = \int f_{XY}(x, y)dx \tag{60}$$

Let X and Y be two continuous random variables with joint PDF $F_{XY}(x, y)$. Assume $f_X(x) > 0$, then:

$$f_{Y|X=x}(y) = \frac{f_{XY}(x, y)}{f_X(x)} \tag{61}$$

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similarly, if $f_Y(y) > 0$:

$$f_{X|Y=y}(x) = \frac{f_{XY}(x,y)}{f_Y(y)} \tag{62}$$

as the continuous CDF is actually a proper PDF,

$$\int f_{Y|X=x}(y)dy = 1 \tag{63}$$

and, for any set $B \subseteq \mathbb{R}$:

$$P(Y \in B|X = x) = \int_B f_{Y|X=x}(y)dy \tag{64}$$

2.6.4 Properties of Random Variables

Definition 2.6.4.1: Let X and Y be two random variables (if discrete, with values in R and S; if continuous, with values in $\mathbb{R}$). We say X and Y are independent if one of the following equivalent statements hold:

- $f_X(x)f_Y(y) = f_{XY}(x,y)$ for all $x, y \in R \times S$ (discrete), or $x, y \in \mathbb{R}^2$ (continuous).
- $f_{Y|X=x}(y) = f_Y(y)$ for all $x, y \in R \times S$ (discrete), or $x, y \in \mathbb{R}^2$ (continuous) with $f_X(x) > 0$.
- $f_{X|Y=y}(x) = f_X(x)$ for all $x, y \in R \times S$ (discrete), or $x, y \in \mathbb{R}^2$ (continuous) with $f_Y(y) > 0$.
- $P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$ for any two sets A and B.

Definition 2.6.4.2: Suppose X and Y are discrete random variables taking values in R and S respectively. Let $f_{XY}(x,y)$ be the PMF and $f_X(x)$ and $f_Y(y)$ be the marginal PMFs of X and Y. We can define the means and variances as:

$$E(X) = \mu_X = \sum_{x \in R} x f_X(x) \tag{65}$$

$$E(Y) = \mu_Y = \sum_{y \in S} y f_Y(y) \tag{66}$$

$$V(X) = \sum_{x \in R} x^2 f_X(x) - (E(X))^2 \tag{67}$$

$$V(Y) = \sum_{y \in S} y^2 f_Y(y) - (E(Y))^2 \tag{68}$$

Definition 2.6.4.3: The covariance between two discrete or continuous random variables X and Y is denoted by $cov(X,Y)$ and is given by:

$$cov(X,Y) = \sigma_{XY} = E[XY] - E[X]E[Y] \tag{69}$$

You can also compute covariance by this formula:

$$cov(X,Y) = E[(X - \mu_x)(Y - \mu_y)] \tag{70}$$

Similarly, we can measure the correlation coefficient between two variables using:

$$Corr(X,Y) = \rho_{XY} = \frac{cov(X,Y)}{\sqrt{V(X)V(Y)}} \tag{71}$$

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2.7 Correlations

2.8 Central Limit Theorem and Sampling

2.9 Hypothesis Testing

2.10 Data Types and Visualisation

2.11 Basic Data Analysis with Python

3 Data Cleaning and Analytics

This section is taught by Dr Igor Razgon. The recommended reading for this topic is:

- Python for Data Analytics (McKinney, 2022).

3.1 Data Quality and How to Produce High-Quality Data

3.1.1 Data Quality

Definition 3.1.1.1: Data cleansing is the process of detecting and correcting (or removing) corrupt or inaccurate records from a record set, table, or database and refers to identifying incomplete, inaccurate, or irrelevant parts of the data and then replacing, modifying, or deleting the dirty or coarse data.

For data to be considered high-quality, it should meet the following criteria:

- **Definition 3.1.1.2**: Validity: The degree to which the measures conform to defined business rules or constraints.
- **Definition 3.1.1.3**: Accuracy: The degree of conformity of a measure to a standard or a true value.
- **Definition 3.1.1.4**: Completeness: The degree to which all required measures are known.
- **Definition 3.1.1.5**: Consistency: The degree to which the data is consistent, within the same data set or across multiple data sets.
- **Definition 3.1.1.6**: Uniformity: The degree to which the data is specified using the same unit of measure.

3.1.2 How to Produce High-Quality Data

Producing high-quality data is an iterative process moving through inspection, cleaning, verifying, and reporting.

Definition 3.1.2.1: Inspection: Detect unexpected, incorrect, and inconsistent data.

- **Definition 3.1.2.2**: Data profiling is the process of viewing the summary statistics of your data to identify standards or patterns in columns of data. Here it is also important to note the data type, any missing values, the number of unique values, the distribution of values, and the relationship between the data and other columns.
- Visualising the data can also be helpful for identifying outliers, as can using statistical methods such as mean, standard deviation, range, and/or quartiles.

Definition 3.1.2.3: Cleaning: Fix or remove the anomalies detected.

- It is important to identify and remove duplicate entries in your data sets.
- When converting the data into a consistent data type, values unable to fit the format should be flagged and converted to NA value.

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- Beyond the data types, the data should be further standardised. This means strings are of a consistent capitalisation, numerical values use the same measurement unit, and dates are in the same format (pick either the UK or US format).
- It can also be useful to scale/transform your data to fit within a certain scale (i.e. 0-100 or 0-1). This can make data easier to plot (e.g., exponential data being logged).
- You can either replace missing values by using a mask to indicate where they are missing, or choosing a sentinel value to replace the missing entries. Conversely, you can also drop the whole record from the data set, or where possible calculate the missing values from other observations.
- **Definition 3.1.2.4**: We define outliers as values which fall 1.5 times the interquartile range from both the first and third quartiles.

Definition 3.1.2.5: Verifying: After cleaning, the results are inspected to verify correctness.

Definition 3.1.2.6: Reporting: A report about the changes made and the quality of the currently stored data is recorded.

3.2 Working with Files

There is no new content here, it is just an introduction to Python's file read/write operations.

3.3 Pandas and Numpy

There is no new content here, it is just an introduction to Pandas and Numpy.

3.4 Data Cleaning and Preparation

3.4.1 Missing Values and Duplicates

See notes from Artificial Intelligence.

3.4.2 Removal of Outliers

Definition 3.4.2.1: When trying to identify outliers, we need to provide precise specifications for a rare occurrence. For this, we generally use one of three interpretations:

- *Threshold Limit*: We set some threshold limit, say 3, and all values that occur fewer than 3 times are considered outliers.
- *Threshold Limit for Clusters*: We classify all the values into several classes and again set a limit, say 3. A value is considered an outlier if its class occurs fewer than 3 times.
- *Quartiles-Based*: We apply a standard statistical methodology involving quartiles and interquartile range.

3.4.3 Splitting and Merging

See notes from Artificial Intelligence.

3.5 Python Data Visualisation Libraries

3.6 Data Analytics and Reporting

4 Computer Graphics

This section is taught by Dr George Alex Koulieris. Check for recommended reading.

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5 Image Processing

This section is taught by Dr George Alex Koulieris. Check for recommended reading.

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Part I
Appendix

6 Bibliography

6.1 Module Professors

Module Lead: Professor Thomas Erlebach

- Focus: Algorithms and networks.
- Email: thomas.erlebach@durham.ac.uk
- Office Hours: n/a.

Lecturer: Doctor Igor Razgon

- Focus: Algorithms and complexity, graph theory, and discrete mathematics.
- Email: igor.razgon@durham.ac.uk
- Office Hours: Friday 14:00 - 15:00.

Lecturer: Doctor George Alex Koulieris

- Focus: Imagr processing and computer graphics.
- Email: georgios.a.koulieris@durham.ac.uk
- Office Hours: Please email to make an appointment.

6.2 References

6.3 List of Figures

6.4 Reading List

6.4.1 Probability

Degroot, M.H. and Schervish, M.J. (2012). Probability and Statistics. Boston Mass.: Pearson.

6.4.2 Data Cleaning and Analytics

McKinney, W. (2022). Python for Data Analysis. O'Reilly Media, Inc.